§3 Evaluating Definite Integrals

∗ <u>The Evaluation Theorem</u>

f : continuous on $[a,\ b]$

$\Rightarrow\ \int_a^b f(x)\,dx = F(b) - F(a)$ where F is any antiderivative of f, that is, $F' = f$.

<u>Notation</u> $F(x)\big]_a^b = F(b) - F(a)$

$\Rightarrow\ \int_a^b f(x)\,dx = F(x)\big]_a^b = F(b) - F(a)$

<u>Exam 1</u> Evaluate $\int_1^3 e^x\,dx$.

(sol) $\int_1^3 e^x\,dx = e^x\big]_1^3$

$= e^3 - e$

<u>Exam 2</u> Find the area under the cosine curve from 0 to b, where $0 \le b \le \pi/2$,

(sol) The area is

$$A = \int_0^b \cos x\,dx = \sin x\big]_0^b = \sin b$$

∗ <u>Indefinite Integral</u>

$\int f(x)\,dx$ is traditionally used for an antiderivative of f and is called an **indefinite integral**.

$\Rightarrow\ \int f(x)\,dx = F(x) + C$ where $F'(x) = f(x)$

<u>Note</u> $\int_a^b f(x)\,dx$ is a number, but $\int f(x)\,dx$ is a function of x.

Exam 3 Find the general indefinite integral

$$\int (10x^4 - 2\sec^2 x)\, dx .$$

(sol) $\int (10x^4 - 2\sec^2 x)\, dx = 2x^5 - 2\tan x + C.$

Exam 4 Evaluate $\int_0^3 (x^3 - 6x)\, dx$.

(sol) $$\int_0^3 (x^3 - 6x)\, dx = \left.\frac{1}{4}x^4 - 3x^2\right]_0^3$$
$$= -\frac{27}{4}$$

Exam 5 Find $\int_0^2 \left(2x^3 - 6x + \frac{1}{x^2+1}\right) dx$.

(sol) $$\int_0^2 \left(2x^3 - 6x + \frac{1}{x^2+1}\right) dx = \left.\frac{x^4}{2} - 3x^2 + \tan^{-1}x\right]_0^2$$
$$= 8 - 12 + \tan^{-1}2$$
$$= \tan^{-1}2 - 4$$
$$\approx -0.67855$$

Exam 6 Evaluate $\int_1^9 \frac{2t^2 + t^2\sqrt{t} - 1}{t^2}\, dt$.

(sol) $$\int_1^9 \frac{2t^2 + t^2\sqrt{t} - 1}{t^2}\, dt = \int_9^1 (2 + t^{1/2} - t^{-2})dt$$
$$= \left. 2t + \frac{2}{3}t^{3/2} - \left(-\frac{1}{t}\right)\right]_1^9$$
$$= \frac{292}{9}$$